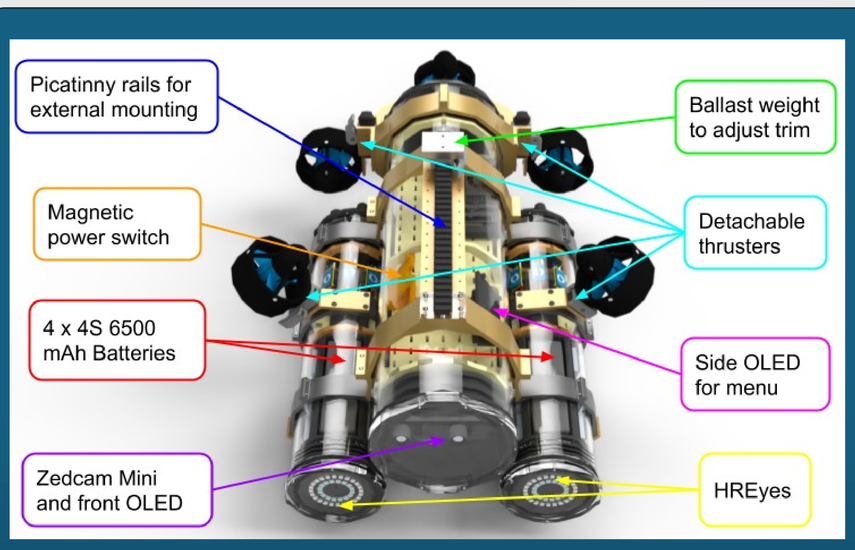
Design and Development of the MeCO Open-Source Autonomous Underwater Vehicle

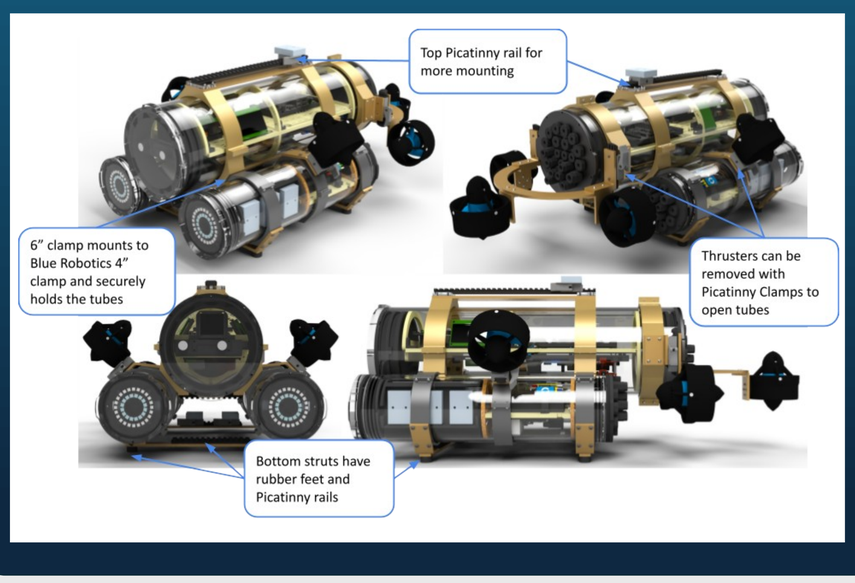
David Widhalm, Cory Ohnsted, Corey Knutson, Demetrious Kutzke, Sakshi Singh, Rishi Mukherjee, Grant Schwidder, Ying-Kun-Wu, and Junaed Sattar

University of Minnesota Twin Cities



- Five thrusters for powerful maneuvering.
- Four batteries – long battery life.
- High-performance onboard computing.

Enough power to tow a diver!



- Modular Picatinny rails for external payload.
- Optional data-only tether.
- Supports hot-swappable tether connection.

MeCO: Medium Cost Open-Source AUV

- Affordable and fully open-source AUV with a base cost of ~ **$10k**.
- Reconfigurable and field-ready, with a modular design that supports rapid adjustments.
- Companian-capable and supports diver missions using two-way communication.
- Offers field proven control along with advanced real-time inference and visual SLAM capabilities.

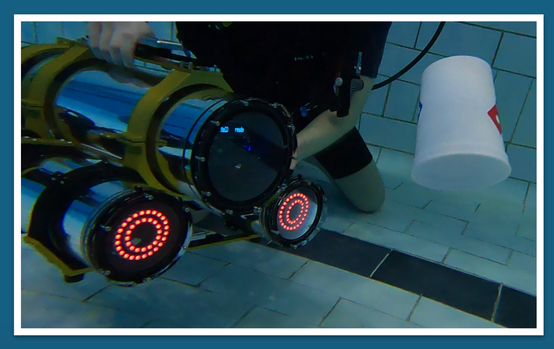
Characteristics

Dimensions (cm)	80 x 52 x 28	Overall
Displacement (L)	23.6	
Buoyant Mass (kg)	23.6 (fresh)	24.2 (salt)
Installed Ballast (kg)	2.7 (fresh)	3.3 (salt)
Battery Capacity (Wh)	385	
Thrusters	5x T-200 (59 N)	Non-Holonomic
Computing	Jetson Orin	4x Teensy 4.1

Audio Visual Feedback

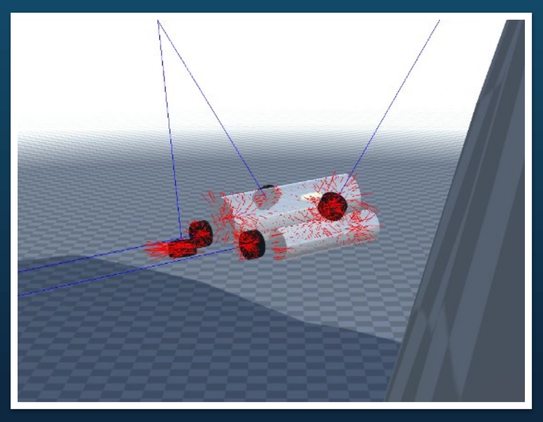


OLED displays are individual controlled and can relay textual information to divers including: status, current function context, selection menus, and warnings.



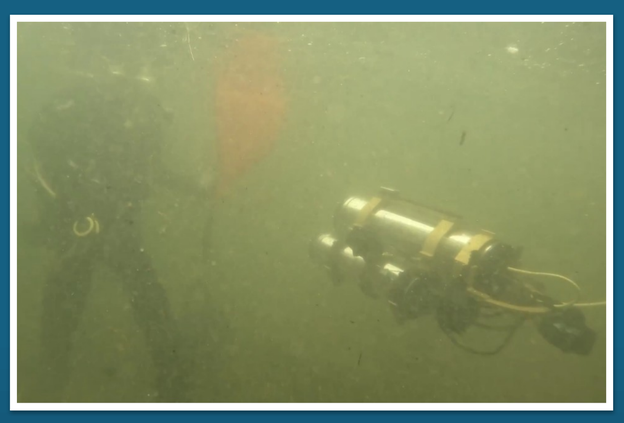
HREye and Siren Systems can provide tailored feedback for any situation, greatly enhancing the interaction between divers and MeCO.

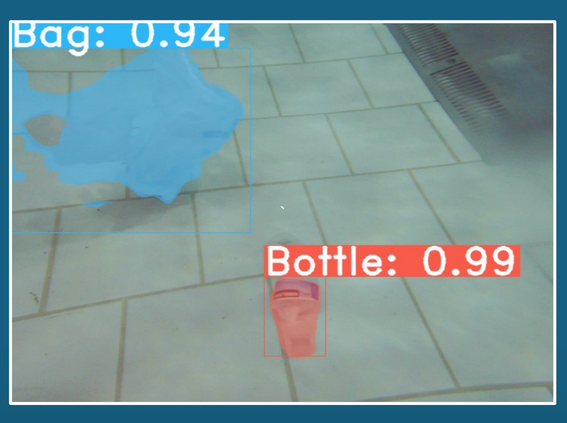
Simulation

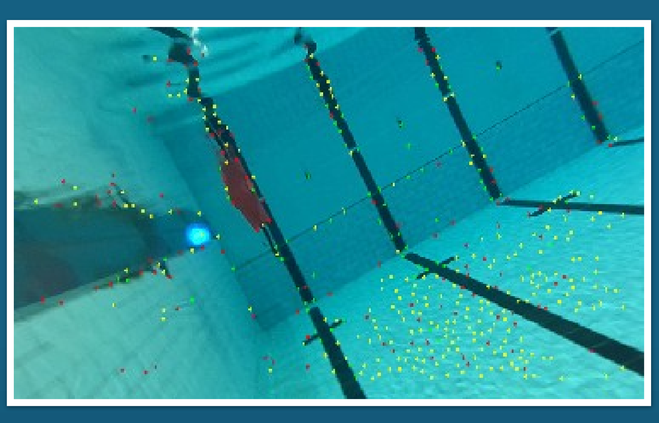


A lightweight real-time simulation has been developed that can be used for system development on MeCO. Internal systems are simulated, and models are physics based so it behaves similarly to the actual robot.

Experiments





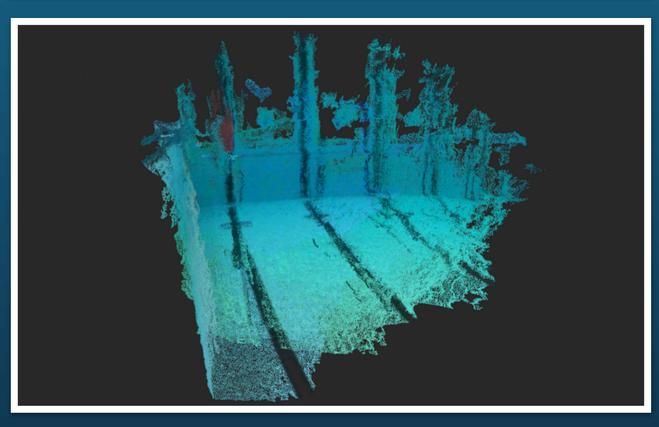




MeCO can be quickly reconfigured for fresh or salt water by adjusting ballast for new control patterns. It has completed fully tetherless open-ocean missions and can be powered-on in-situ to operate without surface support. These capabilities have been tested in diverse field conditions.



The computing resources are powerful enough to run inference allowing MeCO to reason about its surroundings and understand gestures from divers for human-to-robot communication



Visual SLAM can create 3D maps of the underwater environment through stereo camera

IRVLAB

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